Exp No:10 Experiment to understand array function in Data science.

Description: Understand array function using Numpy library

Code:

```
import numpy as np

array = np.random.randint(1,100,9)

array

array([56, 85, 37, 25,  8, 71, 11, 14, 47])

np.sqrt(array)

array([9.48683298, 5.09901951, 6.63324958, 6.63324958, 5.56776436,

    5.47722558, 9.48683298, 9.94987437, 6.4807407 ])


new_array = array.reshape(3,3)

1


new_array

array([[56, 85, 37],

    [25,  8, 71],

    [11, 14, 47]])


new_array.ndim

2


new_array.ravel()

array([56, 85, 37, 25,  8, 71, 11, 14, 47])


newm = new_array.reshape(3,3)

newm
```

```
array([[56, 85, 37],
       [25,  8, 71],
       [11, 14, 47]])

newm[2,1:3]
array([14, 47])

newm[1:2,1:3]
array([[ 8, 71]])

new_array[0:3,0:0]
array([], shape=(3, 0), dtype=int64)

new_array[0:2,0:1]
array([[56],
       [25]])

new_array[0:3,0:1]
array([[56],
       [25],
       [11]])

new_array[1:3]
array([[25,  8, 71],
       [11, 14, 47]])
```

Result:

Thus Understand array function using Numpy library